The University of Auckland
Faculty of Engineering
Department of Software Engineering

Part IV Research Project Final Report

A Smart Inventory Management System with Asset Tracking

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ABSTRACT

The demand for efficient indoor localization systems has increased as computational technology advances. Yet, many university technicians still need to rely on outdated methods, such as manual spreadsheets, to manage lab inventories, which can lead to errors and delays in operation. While adequate for outdoor usage, traditional GPS solutions are ineffective for indoor environments due to signal attenuation, highlighting the need for a more advanced asset management system. This research addresses these limitations by evaluating the suitability of Ultra-Wideband (UWB) and Bluetooth Low Energy (BLE) technology for tracking and managing laboratory equipment within the ECSE department at the University of Auckland. The study assesses the viability of a prototype system by analyzing its accuracy and responsiveness in detecting asset location at the room level. By conducting a comparative analysis of UWB and BLE technologies along with the user feedback requirements, the prototype will demonstrate how an improved asset-tracking system can enhance operational efficiency in managing research equipment in university labs.

DECLARATION

Student : Youngmin Kim

I hereby declare that:

1. This report is the result of the final year project work carried out by my project partner Devesh Duptala and I, Youngmin Kim under the guidance of our supervisor Duleepa Thrimawithana in the 2024 academic year at the Department of Electrical, Computer and Software Engineering, Faculty of Engineering, University of Auckland.
2. This report is not the outcome of work done previously.
3. This report is not the outcome of work done in collaboration, except that with a potential project sponsor (if any) as stated in the text.
4. This report is not the same as any report, thesis, conference article or journal paper, or any other publication or unpublished work in any format.

In the case of a continuing project, please state clearly what has been developed during the project and what was available from previous year(s):

This is a new project built this year in 2024.

Signature: Youngmin Kim

Date: 13/10/2024

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Finally, I want to give warm thanks to my family. While they were not directly involved in this project, their emotional support and encouragement motivated me to push forward and complete my journey through software engineering.

Glossary of Terms

Term	Definition

Abbreviations

ILS	Indoor Localization System
TN	Target Nodes
AN	Anchor Nodes
RSS	Received Signal Strength
TOA	Time of Arrival
AOA	Angle of Arrival
RF	Radio Frequency
RSSI	Received Signal Strength Indicator
AP	Access Points
RFID	Radio Frequency Identification
BLE	Bluetooth Low Energy
UWB	Ultra-Wideband
INS	Inertial Navigation Systems
TOF	Time-of-Flight
TWR	Two-Way Ranging
NLOS	Non-line-of-Sight
LOS	Line-of-Sight
R^2	Coefficient of Determination

1. Introduction

This report outlines the intention behind our project, which focuses on researching an accurate indoor localization system for the university lab environment, specifically within the ECSE department. Titled "A Smart Inventory Management System with Asset Tracking," the project explores identifying high-accuracy, cost-effective, and user-friendly technologies to track and manage assets in indoor environments efficiently.

For my Part IV Final Year research paper in the Bachelor of Software Engineering (Hons), I have worked alongside my partner, Devesh Duptala, under the guidance of our supervisor, Professor Duleepa Thrimathwana. Our research has been conducted within the lab environment across Level 7 laboratories. The motivation for this project arises from the everyday challenge of locating assets in various settings. Initially, we considered a smart system for asset management that would be particularly useful in asset-concentrated environments, such as shops or stores. Our supervisor then expanded on this idea, recognizing that the same issue exists within university laboratories, making this an ideal setting for our research.

1.1. Literature Review

1.1.1. Indoor Localization System

The Indoor Localization System (ILS) is used to determine the location of objects within a closed indoor environment. It applies across various environments such as surveillance, warehouses, healthcare, laboratories, and airports [1]. This system consists of two primary elements: Reference points and Target points. Reference points are stationary points placed at predetermined locations within the designated area. In contrast, Target points are attached to or carried by the objects or individuals needing localization [1]. ILS can function using specialized devices or standard hardware, with the localization processor typically operated as a cloud service. However, it can also run on local servers or mobile devices [2]. Upon user requests, this process calculates locations and ideally displays them back on a digital map to enhance accuracy and provide context to the location data [2]. The operation of an ILS unfolds in two phases, focusing initially on measuring signal parameters such as RSS between the Target point and multiple Reference points. These measurements are then used to determine the position of the Target relative to the Reference [1]. ILS employs one of two primary localization algorithms to accurately estimate the target's location based on these signal measurements.

1.1.2. Detection Algorithms

Range-based algorithms in indoor localization work similarly to GPS, using trilateration techniques [2]. These methods calculate the distance or direction from the TN to multiple fixed anchor nodes ANs based on parameters like RSS, TOA, or AOA [1]. For example, using RSS,

the system estimates the distance by measuring the signal strength between the TN and ANs, where a stronger signal indicates closer proximity. Trilateration is then used to determine the target's exact position by calculating the intersection of circles formed by distance measurements from multiple ANs. In contrast, range-free algorithms, such as fingerprinting, do not calculate distances. Instead, they rely on the unique signal patterns, or "fingerprints," associated with different locations. These fingerprints are collected during a preliminary survey and stored in a fingerprint database [1]. During system operation, real-time signal measurements from the target are compared to this pre-collected database to estimate the location. Fingerprinting can accommodate data from various sources, including RF signals, acoustic signals, images, and magnetic fields [2]. For our application, we used an existing range based Received Signal Strength Indicator (RSSI) method for position estimation due to its simplicity and modern usage [3]. This method is simple and fast and doesn't require specialized infrastructure, making it more practical than other localization methods [4].

1.1.3. Related Study

Existing indoor localization technology varies, including Wi-Fi, RFID, BLE, UWB, Infrared, Satellite, INS, Magnetic, Sound, etc. However, our proposed approach will focus on exploring popular RF indoor localization technologies such as RFID, UWB, BLE, and Wi-Fi. We will develop comparative metrics based on specific criteria to evaluate these technologies.

1.1.4. Wi-Fi

Wi-Fi-based location tracking determines a device's position by measuring signal strength from multiple Wi-Fi APs. By comparing the signal strength from each AP, the system triangulates the device's location, using the variations in signal patterns to estimate its proximity to the APs [5]. This method is widely used due to the ubiquity of Wi-Fi infrastructure in many indoor spaces, making it convenient and cost-effective [6]. However, it has limitations. Wi-Fi localization can suffer from signal attenuation caused by obstacles like walls, leading to reduced accuracy [5]. Nearby APs, such as those from adjacent buildings or hotspots, can interfere with the data collection process due to their weak signal strength and inconsistent availability, as they are designed primarily for communication rather than localization. This interference affects the accuracy of location tracking, which then requires more efficient algorithms to correct the inaccuracies. However, implementing these advanced algorithms introduces deployment of system cost and latency in system [6], [7].

1.1.5. BLE

Bluetooth-based indoor localization uses BLE beacons placed throughout a location to transmit signals that devices can detect to estimate their proximity to the beacons [7]. The system

determines the device's location by identifying the closest beacon, making it effective for room-level or zone-level localization. Technologies like iBeacon use BLE beacons to ping a user's device and estimate their location based on the nearest beacon [8]. BLE is particularly suited for tasks requiring minimal data transfer due to its low latency and smaller data packet size [9]. Compared to Wi-Fi, BLE is more cost-effective, consumes less power, and has a typical range of 70-100 meters, making it ideal for short-range indoor applications such as proximity services and location tracking [9], [10]. However, while BLE is advantageous, its high signal fluctuations due to environmental factors present a challenge in small confined spaces, reducing precise location tracking accuracy [2].

1.1.6. RFID

RFID systems operate through back-scattering communication, where each RFID tag has a unique ID read by RFID readers, which can be managed via a laptop or mobile device [10], [11]. RFID technology is divided into passive and active tags. Active RFID tags have their own power source, allowing them to track items over longer distances (up to 100 meters). In contrast, passive RFID tags rely on power from the reader’s signal, enabling short-range tracking up to 10 meters [10]. RFID offers benefits such as no direct contact, permanent data storage, and strong readability, even in challenging environments [12]. However, it requires expensive readers ($500-$2000) and can lose accuracy when the reader and tag are positioned at sloping angles [13].

1.1.7. UWB

UWB localization uses extremely short pulses spread over a broad frequency spectrum (3.1GHz to 10.6GHz) with a minimal frequency cycle [14]. This method determines the location of a target by measuring the time it takes for these pulses to travel from various anchor nodes to the target by TOF. UWB employs TWR technology to calculate distance based on TOF. The wide bandwidth allows for precise timing of signal arrivals, which is crucial for accurate distance measurements. UWB technology offers very high accuracy, minimal interference, and excellent material penetration [15]. However, in dynamic environments, such as those NLOS conditions, obstacles can block the direct path of the UWB signal. This results in incorrect ToF calculations due to indirect signal paths, leading to ranging errors and potentially inaccurate positioning [15].

1.1.8. Wireless Transmission Technology Analysis for ILS

Our research project evaluates indoor localization methods for a university laboratory, where limited studies have been conducted to address challenges specific to this environment, such as complex layouts and equipment interference. While existing solutions are often designed and tested in open or simplified spaces, a significant gap exists in understanding their effectiveness in

confined, more complicated lab settings. Regarding our project, we focus on four critical criteria: low implementation and operational costs, ease of installation, and high accuracy and precision. Prioritizing cost-effectiveness and minimal disruption, we aim to use an existing system that covers the lab environment adequately. The chosen system must be easy for technicians to adopt, require minimal maintenance, and provide precise location tracking. After reviewing the four well-known RF technologies, Wi-Fi, RFID, UWB, and BLE, we found BLE-based technology, specifically BLE beacons, to be the most practical choice for indoor localization. For moderately sizeable indoor laboratory settings, BLE beacons offer significant advantages. They are cost-effective and easily integrated with personal devices. Additionally, as with Wi-Fi, BLE beacons do not require expensive dedicated readers like RFID or access points for deployment [13]. However, BLE often experiences high RSSI fluctuations caused by environmental obstacles, which reduces its precision, making it less suitable for environments requiring high accuracy [15]. To overcome the limitations of relying on a single technology, fusion positioning technology has emerged to balance accuracy, cost, and power consumption [16]. Fusion technology combines the advantages of different systems to improve performance, and it is becoming a popular trend in indoor positioning. Hence, while our focus is on BLE, we will also consider incorporating a hybrid model with UWB technology that offers greater precision using ToF and TWR methods. This approach aims to enhance accuracy in dynamic environments [16].

1.2. Project Scope and Objectives

Managing assets in university laboratories has become increasingly complex, with the current inventory management system at the University of Auckland relying heavily on manual spreadsheets and SharePoint records. This outdated approach limits operational efficiency and is time-consuming, highlighting the need for an automated inventory tracking system [17]. Inspired by the daily use of GPS for outdoor navigation [18], this project aims to introduce a smart inventory management system that offers ease of use and obtains high accuracy in indoor asset tracking. To achieve this, we will explore the capabilities of BLE and UWB technologies, focusing on their performance, accuracy, and configuration optimization. Our approach includes gathering feedback from key stakeholders, such as ECSE lab technicians, to ensure the system aligns with operational needs. By testing various beacon layouts and configurations, we aim to develop a scalable solution that achieves high accuracy, practical for deployment across university labs.

Project Objectives

1. **Evaluate Indoor Localization Technologies**: Assess the practicality, cost-effectiveness, and accuracy of RF technologies, specifically for BLE and UWB, for real-time asset tracking in indoor environments, identifying the most suitable solution.

2. **Stakeholder Engagement**: Conduct interviews and user surveys with university staff members and technicians to gather insights and ensure that the technology aligns with specific user needs and operational workflows.
3. **Prototype Model Development**: Design and implement a mobile-compatible prototype system using the most suitable model. The research will involve experimenting with different BLE beacon configurations and UWB modules to optimize placement, ensuring maximum localization accuracy while maintaining efficiency and scalability for inventory management.
4. **Iterative Testing and Refinement**: Through iterative pilot testing, we will refine the system by experimenting with different configurations and layouts of BLE and UWB technology. The iterative process will allow for adjustments to be made based on user feedback, improving user experience, operational efficiency, and overall accuracy.

2. System Resources and Tools

2.1. User Survey Research

The user survey gathered insights from 13 participants, primarily PhD students who frequently use lab equipment. The results revealed significant dissatisfaction with the current SharePoint-based asset management system. Notably, 61.6% of respondents reported being neutral or dissatisfied with the system, while 0% expressed complete satisfaction, as Appendix A shows. The primary concerns included the absence of real-time asset tracking, difficulty locating equipment, and a tedious check-in/check-out process. The survey participants strongly preferred a more user-friendly system, with key features such as real-time tracking, easy check-in/out, and asset history logs being the most desired. Specifically, over half of the participants (53.8%) favored an automated system using mobile barcode scanning for equipment management, and 69.2% expressed comfort in adopting new technology. Concerns were raised about the system's reliability, ease of use, and potential learning curve. We understand the importance of a well-designed, simple, intuitive onboarding system. The results reinforce the need for a streamlined, mobile-compatible application that integrates real-time tracking and addresses the current system's inefficiencies.

2.2. Software

Based on the survey results, we decided to implement an indoor localization app on the iOS platform, leveraging the capabilities of Apple's iBeacon technology. iBeacon, created by Apple, allows developers to design applications for short-range environments such as warehouses or laboratories using BLE communication. This technology enables mobile devices to receive location-specific services by detecting nearby BLE signals [19].
For the software, we chose Swift, Apple's programming language for iOS app development, which works seamlessly with Xcode, the Integrated Development Environment (IDE) for

building and deploying iOS apps [20]. Initially, we stored the application data locally using the JSON format, allowing easy data management without requiring an internet connection. However, as the project progressed, we changed to Firebase, a flexible, scalable NoSQL cloud database. Firebase supports real-time data synchronization and offers seamless integration with iOS, making it an ideal choice for our mobile application [21].

2.3. Hardware

We used the ESP32-H2-DevKitM-1-N4 development board for our BLE beacon modules. This kit includes the ESP32 System on Chip (SoC), known for its powerful microcontroller capabilities and support for IEEE 802.15.4, combined with rich peripherals [22]. For the BLE development environment, we utilized the Arduino IDE, an open-source platform, to code, compile, and activate the ESP32 microcontrollers. One key advantage of the ESP32-H2 is its flexibility in broadcasting at a wide range of transmission power levels, from -12 dBm to 9 dBm. This was critical for our experiment, which assessed the differences in RSSI signal strength, accuracy, and range. Once initialized, the software was programmed in Swift and executed via Xcode, where we used the built-in Core Bluetooth framework from ESP32-H2 to receive RSSI signals across our device. For the UWB, we used the DWM3001CDK development kit. This development board integrates the Qorvo DW3110 UWB transceiver and the Nordic nRF52833 SoC, which supports both UWB and BLE functionalities. It has a J-Link debugger, allowing straightforward flashing and programming via USB connections. This kit is compatible with IEEE 802.15.4z BPRF UWB communication, making it ideal for real-time location system (RTLS) applications. Additionally, it integrates smoothly with Apple's Nearby Interaction software, which was crucial for seamless deployment. Both technical details were referenced from the respective product brief documents [23],[24].

3. Experiments and Findings

3.1. Experimental Introduction

This experimental study's primary objective was to evaluate our methodology's feasibility and effectiveness, gather preliminary data, and refine our research objectives and tools. Each experimental session was conducted in a controlled environment with identical settings and was repeated to guarantee the consistency and reliability of the data collected. Further details, including visual documentation of the experimental procedures and setup process, can be found in Appendix B.

3.2. Range and Accuracy Testing of BLE and UWB

In preparation for our experiments, we conducted two tests to assess the minimum and maximum range and accuracy of both UWB and BLE modules. The modules used for these experiments

were the ESP32-H2 for BLE and the DWM3001C for UWB, selected based on price and suitability for our research objectives. The experiment was conducted across the Level 6 and 7 labs of the ECSE department at the University of Auckland. The distance data for UWB was collected using an existing program's built-in app, the Qorvo NI app, while RSSI signals were captured using both a Bluetooth measurement app and our custom-built Swift app for comparison. These initial tests were focused on comparing the measured accuracy and range against theoretical values to evaluate the feasibility of using these technologies for indoor localization and asset tracking in our application.

3.2.1. Accuracy Assessment of BLE and UWB

This experiment aimed to evaluate the accuracy of RSSI signals from BLE and the distance measurements from UWB within an indoor lab environment. Following the methodology outlined in [25]**,** measurements were taken at one-meter intervals for both the BLE and UWB modules, continuing up to a 10-meter interval within one room.

Measurement Setup

The experiment was conducted in three distinct lab environments: open, semi-closed, and closed rooms. To ensure precision, we marked 1-meter intervals starting from one until it reached ten meters on the floor. All doors and windows remained closed during the tests, and no personnel were allowed to enter the area. Both BLE and UWB transmitters were securely taped near the power sources in the lab above the ground to minimize movement interference. After each reading, the system was reset to avoid signal interference. BLE measurements were recorded at two transmission power levels (9 dBm and -12 dBm), with twenty RSSI readings collected at each point. Similarly, for UWB, three distance measurements were recorded at each interval. The process involved two team members: one reading out the recorded values from the system, while the other logged the data into a shared drive for further analysis. The RSSI values were averaged at each point, and the mean values were compared to theoretical RSSI values calculated using the path loss model [26], as shown in Equation 1. For the path loss component (n), we used the value of 3 due to its average value from all different environments 2,3,4, and A was used by the average value of RSSI obtained from a distance of 1m. For UWB, the measured distances were compared with actual values obtained using a physical tape measure. The collected data were further analyzed using excels built in regression analysis's coefficient of determination (R^2) [27], to evaluate the correlation between the measured and theoretical values.

$$\text{RSSI } = -10n\log_{10}(d) + A \tag{1}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{m}(X_i - Y_i)^2}{\sum_{i=1}^{m}(\bar{Y} - Y_i)^2} \tag{2}$$

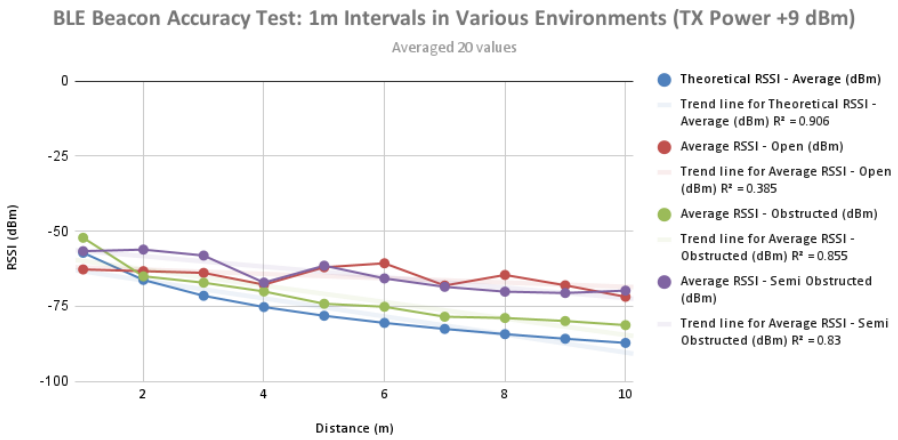


Figure 1: BLE Beacon Accuracy Test for Transmission Power +9 dBm.

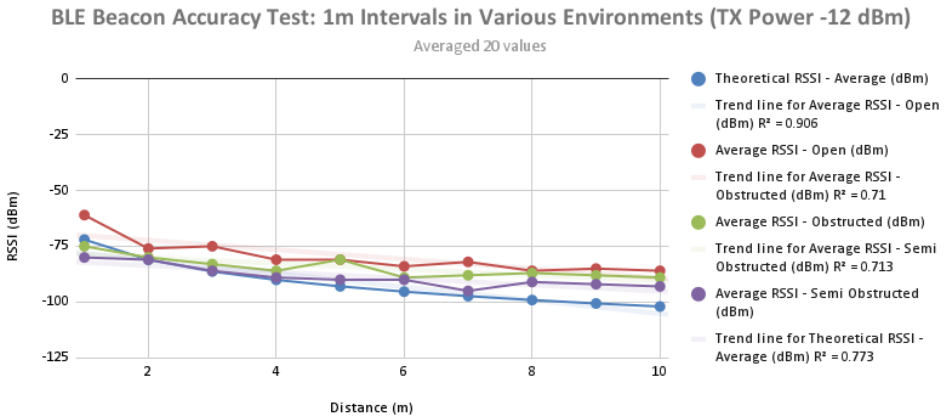


Figure 2: BLE Beacon Accuracy Test for Transmission Power -12 dBm.

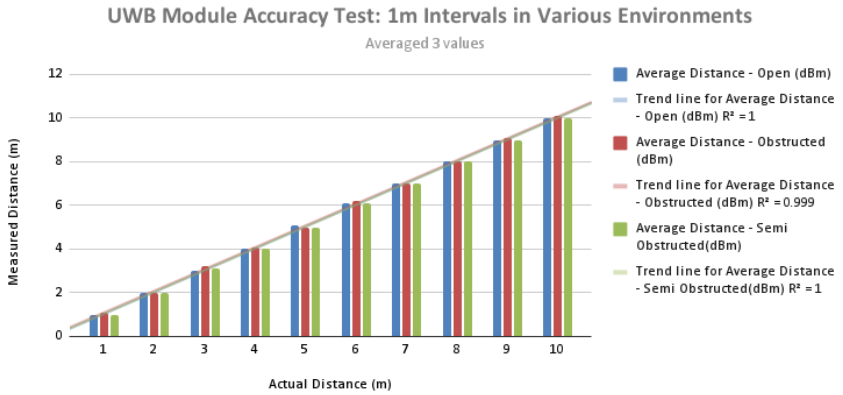


Figure 3: UWB Module Accuracy Test.

3.2.2. Range Assessment of BLE and UWB

This experiment assessed the range and signal strength of both BLE and UWB to determine their maximum and minimum effective range, identifying where the signal remained strong and where it degraded. We followed a methodology based on prior research [28], which utilized the similar DWM1001-DEV module and fixed reference points across the room to estimate proximity using BLE RSSI and UWB range measurements.

Measurement Setup

Measurements were taken at three points in each room: the centre, just before the door, and just after the door (with doors closed). Both BLE and UWB transmitters were securely taped near the power sources in the lab above the ground, positioned close to the centre on the wall, and facing the door at the same angle to ensure a clear LOS for each measurement. We recorded the measurements until the signal was significantly weakened or disconnected. To maintain consistency and minimize the effect of directional changes, the body was kept perpendicular to the transmitter throughout the experiment. BLE readings were recorded at two transmission power levels (9 dBm and -12 dBm), with twenty RSSI values collected at each point. For UWB, three distance measurements were collected at each position. The data collected from this experiment provided insights into the most effective design and implementation for a real-time indoor tracking system.

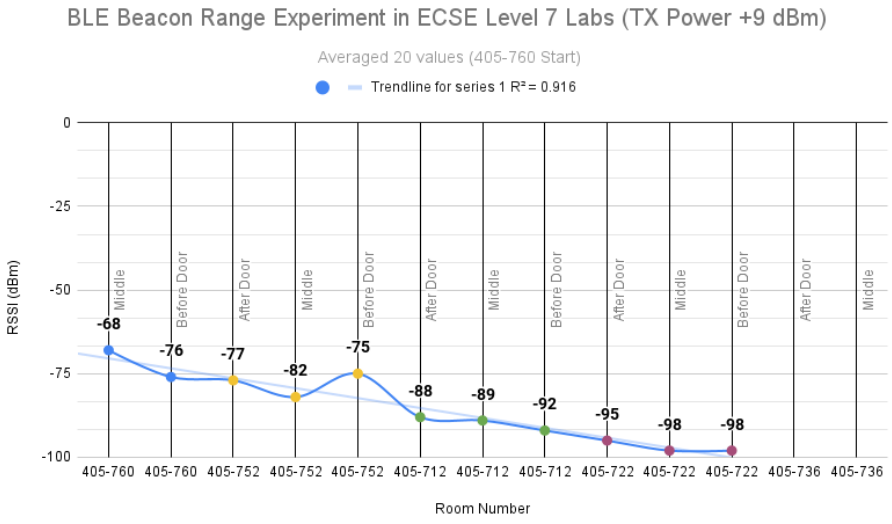


Figure 4: BLE Beacon Range Test for Transmission Power +9 dBm.

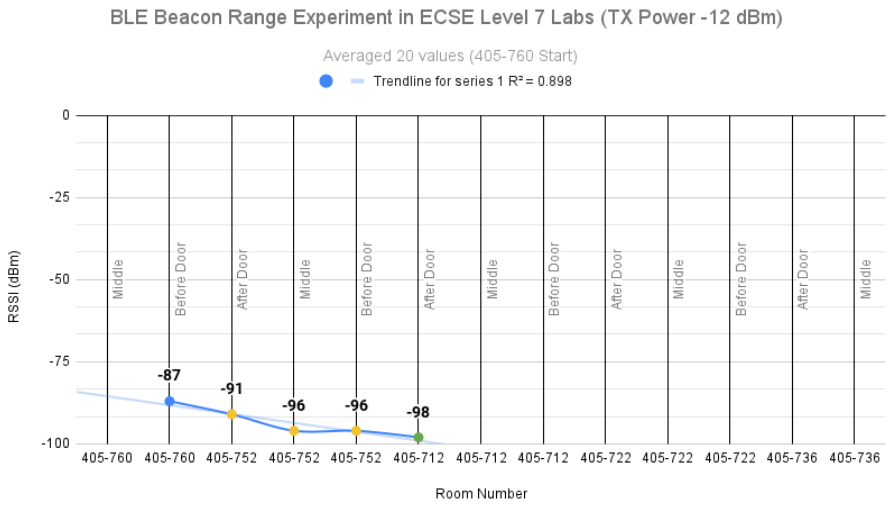


Figure 5: BLE Beacon Range Test for Transmission Power -12 dBm.

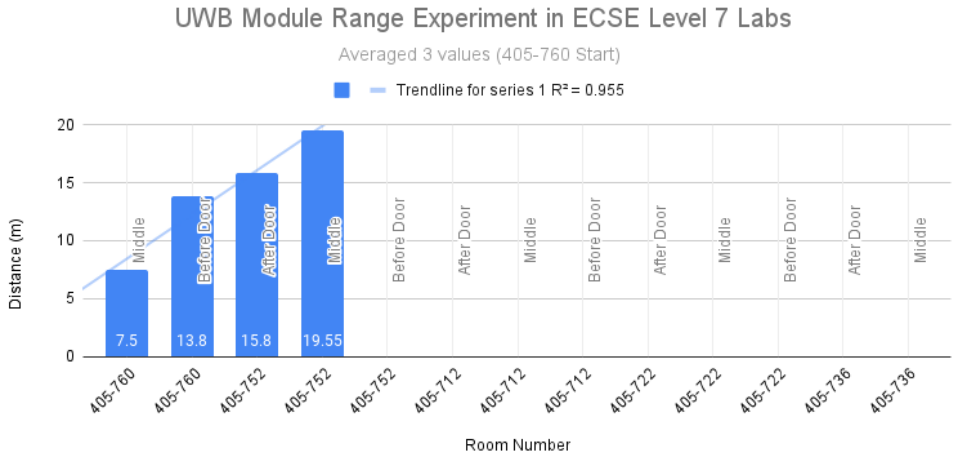


Figure 6: UWB Module Range Test.

3.2.3. Findings

The findings from our experiments show that both BLE and UWB demonstrated varying performance in accuracy and range.

Accuracy: BLE, which relies on RSSI, demonstrated weaker accuracy and greater susceptibility to fluctuations. At a higher transmit power of 9 dBm, BLE experienced a significant drop in accuracy in open environments, with the weakest R^2 value of 0.385. However, the higher initial signal strength allowed BLE signals to remain stable in fully obstructed environments, with an

R^2 value of 0.855. In contrast, at a lower transmit power of -12 dBm, BLE performed better at open environment, with R^2 value of 0.71. However, due to the lower initial signal strength, the signal degraded more quickly over distance, making it less accurate for long-range tracking. In all cases, the R^2 values were below the theoretical R^2 value of 0.906. UWB, on the other hand, maintained near-perfect accuracy across all environments, with an R^2 close to 1, demonstrating minimal deviation from actual values.

Range: At a higher transmit power of 9 dBm, BLE performed reliably over longer distances, up to 35-40 meters, making it better suited for larger areas. However, the RSSI signals weakened heavily beyond specific points like Room 722, leading to challenges in maintaining consistent signal readings. When the transmit power was reduced to -12 dBm, BLE's range significantly decreased, with signal drops occurring as early as Room 752, around 15-20 meters away. UWB exhibited excellent accuracy over short ranges but experienced signal degradation after moving into further rooms, particularly Room 752. Materials like glass or coated walls also caused significant signal attenuation for UWB.

In these experiments, UWB proved superior in providing high accuracy over short distances. BLE, especially at higher transmit power (9 dBm), was more effective for long-range coverage and obstructed environments. BLE at lower transmit power (-12 dBm) performed better in short-range settings at open space but struggled to maintain consistent accuracy. In both cases, RSSI accuracy fluctuated and was influenced by obstacles, signal degradation, and environmental changes.

3.3. RSSI Pattern Analysis & BLE Beacon Configuration

Our experiment initially found that using a single BLE beacon across lab floor seven was insufficient for accurate indoor localization. The range of a single BLE beacon didn't provide total coverage across the entire floor, and the RSSI signal accuracy was not precise enough to consistently distinguish room changes. While the UWB module provided good accuracy in measuring distance, it struggled with detecting directional changes, making it challenging to pinpoint transition points near doorways. Additionally, the high cost and limited range of UWB made it less suitable for large-scale deployments. So, to improve room-level detection, we configured the system with multiple BLE beacons by placing one beacon per room rather than relying on a single beacon to differentiate every room. We aimed to test whether increasing the number of BLE beacons would enhance the system's ability to differentiate the room changes. By analyzing multiple RSSI patterns between adjacent beacons, we sought to determine if this would more effectively distinguish room boundaries. Based on our previous experiment, we found that higher transmission power helped mitigate issues related to signal strength degradation over longer distances, which are essential for refining the accuracy of room-level localization across different environments [29]. We expect this multi-beacon setup to capture more distinct RSSI variations, leading to more precise and reliable room detection.

3.3.1. RSSI Pattern Experiment: Room Boundary Detection with Two BLE Beacons.

Experiment

Based on the limitations identified with a single BLE beacon, we expanded the setup to include two BLE beacons to improve room transition accuracy and minimize RSSI signal ambiguity. One beacon was placed in Room 760 and the other in Room 736, positioned at opposite ends of the floor. Both beacons were configured to broadcast at a transmission power of +9 dBm to maximize their signal range and strength. The aim of the experiment was to determine whether the coverage of two BLE beacons could span the entire floor, ensuring that all rooms were captured. We also wanted to see if there would be noticeable signal patterns between the beacons, which would reduce the ambiguity caused by relying on a single beacon. This overlap was expected to improve room transition detection and allow for better differentiation between adjacent rooms. To assess this, measurements were taken at three specific points within each room: the middle of the room, just before the door, and immediately after the door. At each point, the RSSI values from both beacons were recorded to evaluate signal strength and the degree of overlap. Each measurement was reset between intervals to prevent interference. Data was collected over several intervals, and the results were averaged to provide a reliable set of RSSI values for analysis.

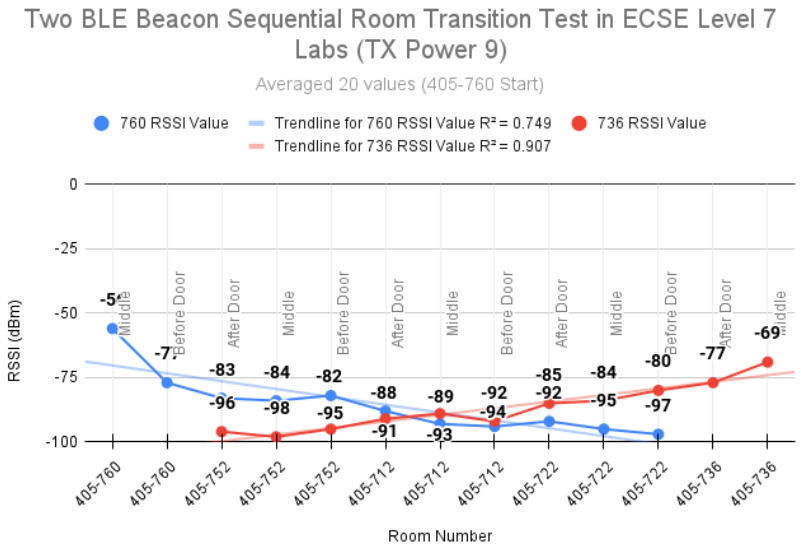


Figure 7: RSSI Pattern Experiment with two BLE beacon.

Findings

The experiment's results in Figure 7 showed that using two BLE beacons improved overall signal coverage, but it clearly failed to establish room boundaries as expected. Initially, we hypothesized that multiple RSSI values would show a noticeable pattern to indicate room

transitions. However, the results revealed that while RSSI changes were distinct near the beacon (within a few meters), the signal variation became minimal as the distance increased.
For example, near the beacon in Room 760, strong RSSI values in the -50s were observed, providing a clear signal. However, as we approached the door to Room 752, the signal sharply dropped near -80s. Once inside Room 752, the RSSI only slightly decreased, ranging from -80 to -84, which made it difficult to distinguish the transition between these two rooms based on signal strength alone. Similarly, the second beacon in Room 736 became slightly detectable in Room 752, but its weak signal made identifying transitions between rooms challenging. In Room 712, the signal from Room 760 further weakened to between -88 and -94, but this change was still too limited to define room boundaries clearly. The signal from Room 736 remained weak, ranging from -89 to -92. By the time we reached Room 722, the signal from Room 760 was almost undetectable, while the signal from Room 736 showed slight variation from what was observed in Room 712, making room transitions challenging to detect. Although there were some noticeable changes in RSSI at shorter ranges, these differences became negligible as the signal travelled further from the source, especially in the middle of the floor.

3.3.2. RSSI Pattern Experiment: Improving Room-Level Distinction Using Three BLE Beacons

Experiment

To address the difficulty of defining central rooms with only two beacons, we introduced a third BLE beacon in Room 712. This beacon was added to improve room-level distinction by covering the central portion of the floor. The experiment followed the same procedure as earlier tests, with measurements taken at key locations within each room. We expected that the third beacon would improve coverage and reduce the effects of RSSI fluctuations between adjacent rooms.

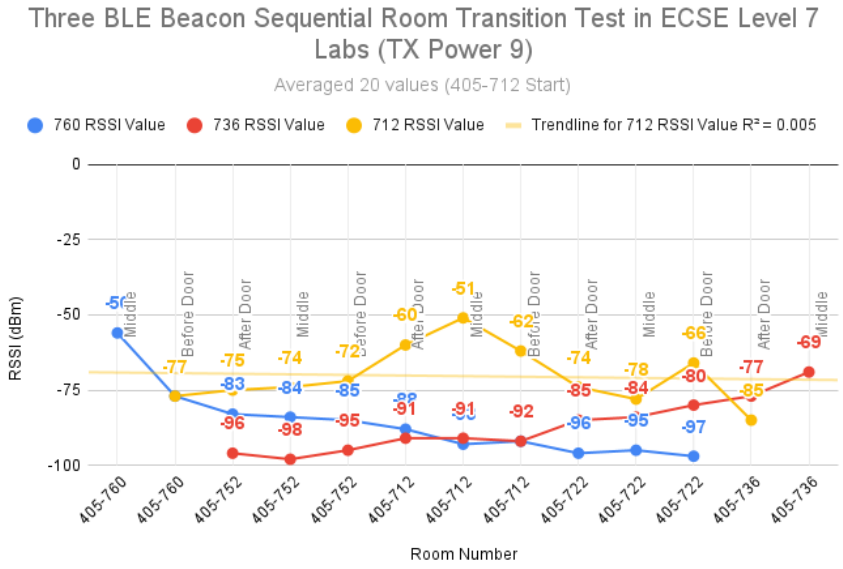


Figure 8: RSSI Pattern Experiment with three BLE beacon.

Findings

Figure 8 shows improved performance compared to the two-beacon configuration. In Room 760, the signal from the beacon in Room 760 provided strong coverage with RSSI values at -50, while the other beacons were undetectable. Moving towards Room 752, the beacon in Room 712 started providing a stronger signal with an RSSI value of around -70 to -80, and the beacon in Room 760 became weaker, dropping to around -85. In Room 712, the beacon provided the strongest coverage, with values around -60, while the beacons in Rooms 736 and 760 had much weaker signals. This strong signal from Room 712 allowed for better distinction in this central region. However, the experiment also revealed new findings. The findings showed that rooms without dedicated beacons, such as Rooms 722 and 752, still experienced issues with boundary detection. RSSI signals from Rooms 712 and 736 or 712 and 760 overlapped, but this overlap was insufficient to separate the rooms and their adjacent areas. Despite the improvements in distinguishing the central room's location, the signal strength remained insufficient for accurate room-level distinction in these spaces.

3.3.3. RSSI Pattern Experiment: Maximizing Signal Coverage with Five BLE Beacons

Experiment

In this experiment, we deployed all five BLE beacons across each room to test whether this configuration could improve room-level accuracy and resolve the ambiguity in room transitions observed in previous tests. The beacons were placed at the center of each room to ensure uniform signal distribution in all directions. The goal is to determine whether placing one beacon in each room will result in more explicit room boundaries and reduce the signal overlap in distinguishing between adjacent rooms. This approach is expected to minimize ambiguity in signal transitions and improve the accuracy of room detection. RSSI values were recorded as users moved through different rooms, and changes in signal strength patterns at room boundaries were observed and analyzed.

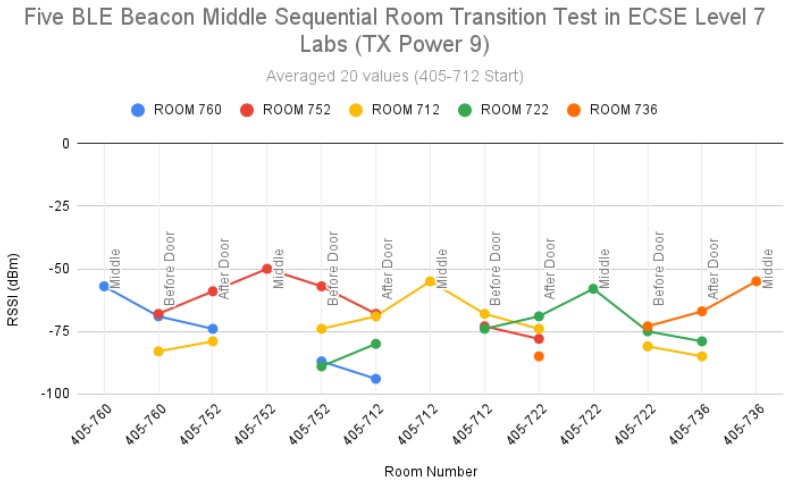


Figure 9: RSSI Pattern Experiment with five BLE beacon.

Findings

Overall, the five-beacon configuration improved room change detection primarily because the increased density of beacons provided more signal data points, leading to more accurate tracking of room change as shown in figure 9. With beacons placed closer together, the RSSI signals were stronger and more reliable when we arrived at the rooms. However, despite these improvements, relying solely on RSSI signal patterns proved inadequate for accurate room detection due to the constant fluctuations in RSSI values. These fluctuations, combined with varying sizes and layouts of rooms, caused the signal strength to weaken unpredictably. In addition, we found that while the central region of each room, where the beacon was placed, had more stable signals, peripheral areas like transition points or room edges were less reliably detectable. This led to challenges in detecting room changes, as transitions often occurred while users were still within the room, with signal reliability only stabilizing near the beacon's central location.

3.3.4. BLE Configuration Experiment – Four Beacons for Enhanced Room Detection

Experiment

After analyzing the results from the RSSI pattern analysis experiments, we identified that while the increased density of beacons enhanced accuracy of room location, relying solely on RSSI pattern still didn't provide accurate room differentiation, especially during transition point. To solve this issue, we focused on improving room level boundary detection by capturing the two strongest RSSI signals for localization, rather than depending on multiple RSSI signal pattern change. For optimal beacon placement, we mounted two BLE beacons above the doors between adjacent rooms, ensuring a clear line of sight to minimize signal obstruction. The experiment was conducted across room 752 to 736. This setup aimed to capture stronger signals from multiple beacons during room transitions, reducing the impact of RSSI fluctuations caused by relying on a single signal. We also expect to have improved room boundary detection, resolving the issues we encountered when the beacons were placed in the center of the room.

Room Number	Location	Beacon 1	Beacon 2	Beacon 3	Beacon 4
405-752	After Door	-59	-74	-84	
405-752	Middle	-66	-66	-82	-90
405-752	Before Door	-69	-63	-80	-89
405-712	After Door	-75	-61	-67	-84
405-712	Middle	-79	-69	-69	-83
405-712	Before Door	-82	-72	-60	-86
405-722	After Door		-81	-60	-79
405-722	Middle		-81	-70	-70
405-722	Before Door		-84	-74	-64

Figure 10: BLE Configuration Experiment with four BLE beacons.

Findings

As detailed in Figure 10, the results confirmed that capturing the two strongest RSSI signals was more effective than relying on RSSI pattern changes. This approach provided clearer room

boundaries, especially in smaller rooms like 752 and 712, which significantly benefited. By focusing on two signals, we reduced the impact of fluctuating RSSI patterns, particularly in smaller rooms like 752 and 712, improving room-level distinction and resolving boundary ambiguity present in previous experiments. To validate the solution's reliability, we further tested the system's accuracy. Following the method from [30], we divided the lab into small sections and tested the system's ability to locate the user in each section. After waiting 10 seconds for each reading, measurements were recorded. Accuracy was calculated using the formula below:

$$Accuracy = \frac{Correct\ Prediction}{Total\ Prediction}\% \quad (3)$$

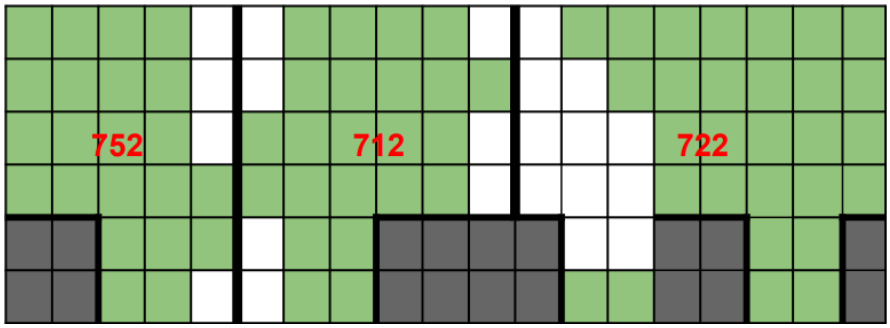


Figure 11: Accuracy Matrix of Four Wall Mounted Beacons test.

The system achieved a room-change detection accuracy of 77.1%, with 74 correct predictions out of 96 attempts, showing an improvement over the previous five BLE beacon setup. In the earlier configuration, signal coverage and RSSI fluctuations were mainly limited to the beacon's central region. This new setup expanded the coverage and reduced the impact of RSSI fluctuation by relying on more stable signal points. However, this accuracy level is still below the desired precision for room-level localization. This issue was particularly noticeable when adjacent rooms were of different sizes. In such cases, ambiguity occurred in defining room boundaries. For example, during transitions from a smaller room like 712 to a larger room like 722, the two strongest signals often came from the closer, smaller room. In an ideal scenario, Room 722 should detect signals from 722 and 736 to determine its location. However, in practice, the signal from Room 712 was often stronger than from Room 736 due to proximity and confusing room transitions. As a result, the system mistakenly attributed the stronger signals from the smaller room, overshadowing those from the larger room and leading to inaccurate room boundary detections.

3.3.5. BLE Configuration Experiment – Four Beacons with Auxiliary Setup for Improved Room Detection

Experiment

Building on the findings from the previous experiment, this test aimed to further improve the

accuracy of BLE signals by addressing the limitations of signal overlap caused by room size differences, particularly when transitioning from smaller rooms to larger rooms. In earlier RSSI pattern experiments, we observed that increasing the number of beacons helped stabilize the overall RSSI readings due to the addition of more signal points. Large rooms, such as Room 722, previously presented signal reliability challenges. To address this, we increased the number of beacons in crucial lab areas, focusing specifically on these larger spaces where the previous setups had struggled. By adding more beacons within the larger room, the system can more effectively capture signals from nearby beacons, reducing the likelihood of incorrectly detecting signals from adjacent rooms and thus improving room-level accuracy. This adjustment is expected to provide a more consistent and accurate differentiation of rooms, especially in spaces where room size had previously led to inaccuracies in boundary detection.

Room Number	Location	Beacon 1	Beacon 2	Beacon 3	Beacon 4	Beacon 5	Beacon 6
405-712	After Door	-56	-75	-84	-86	-82	-90
405-712	Middle	-58	-69	-81	-80	-82	-93
405-712	Before Door	-65	-68	-75	-78	-77	-87
405-722	After Door	-73	-61	-69	-71	-74	-81
405-722	Middle	-75	-75	-61	-63	-71	-84
405-722	Before Door	-80	-69	-71	-68	-55	-75
405-736	After Door	-84	-82	-81	-77	-54	-71
405-736	Middle	-87	-90	-89	-82	-69	-60

Figure 12: BLE Configuration Experiment with six BLE beacons.

Findings

The increased number of beacons significantly enhanced room differentiation, particularly in larger rooms like 722, as illustrated in Figure 12. Boundary detection had been previously unclear in these rooms, but the added beacons provided more precise and reliable signal separation between adjacent spaces. For example, in Room 722, the RSSI values were distinct enough to separate it from both 712 and 736, even when approached near the door, where earlier experiments had shown difficulty. Similarly, Room 712 demonstrated better signal consistency, where the two strongest RSSI signals were kept detected from their own rooms.

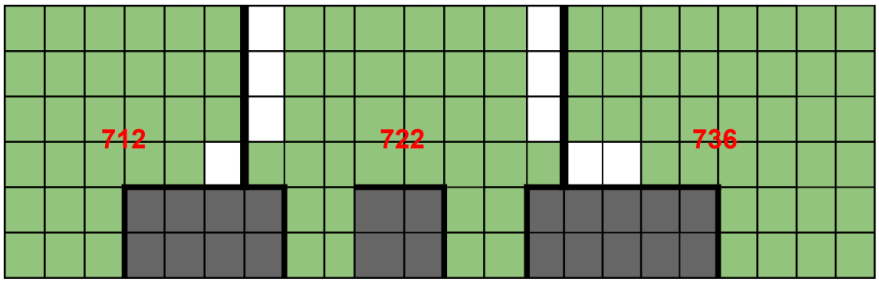


Figure 13: Accuracy Matrix of Four Wall Mounted with Auxiliary Beacons test.

The system achieved a higher room-change detection accuracy of 91.7%, with 101 correct predictions out of 110 attempts. However, minor challenges persisted in accurately defining precise room boundaries. Due to fluctuations in RSSI values, there was a delay as the system averaged the signals before determining the user's location. This lag prevented the system from reaching 100% accuracy. However, despite this limitation, the results demonstrated significant improvements in room-level differentiation and its viability for practical usage.

3.3.6. *BLE Localization Experiment - Quadrant Setup for Improved Localization Accuracy*

Experiment

Building on the successful room-level accuracy of previous BLE beacon configurations, this experiment aimed to refine localization further by dividing the space into quadrants for improved precision. Room 405-722 was split into four quadrants (Upper Left, Upper Right, Bottom Left, and Bottom Right), and five BLE beacons were strategically placed to cover these areas. The hypothesis was that each beacon would provide stronger signals within its designated quadrant, helping to distinguish more effectively between different areas of the room and enhancing overall localization accuracy. The main objective was determining if a quadrant-based setup could offer more precise inner-room localization. RSSI values were collected at the centre of each quadrant and at key points, as well as the room entry and exit, to evaluate the system's ability to identify the quadrant where the user was located correctly.

Room Number	Location	Beacon 1	Beacon 2	Beacon 3	Beacon 4	Beacon 5	Expected	Actual	Accuracy
405-722	Bottom Left Qua	-75	-68	-65	-78	-52	1,3,5	2,3,5	Not accurate
405-722	Bottom Right Qu	-87	-65	-71	-73	-71	3,4,5	2,3,5	Not accurate
405-722	Upper Left Quad	-66	-69	-64	-83	-66	1,2,3	1,3,5	Not accurate
405-722	Upper Right Qua	-79	-56	-73	-74	-71	2,3,4	2,3,5	Not accurate
405-722	Room entry	-66	-60	-66	-83	-71	1,3,5	1,2,3	Not accurate
405-722	Before room exit	-77	-60	-68	-64	-71	3,4,5	2,3,4	Not accurate

Figure 14: Inner-Room Localization Accuracy Test Using Multi-Beacon Configuration.

Findings

The results from Figure 14 showed that the quadrant setup did not perform the anticipated improvement in localization accuracy. Across all quadrants, the detected signals did not consistently align with the expected signals from the nearest beacons. For example, in the Bottom Left quadrant, it was predicted that beacons 1, 3, and 5 would provide the most vital signals; however, the system instead detected signals from beacons 2, 3, and 5, resulting in inaccurate positioning. This inconsistency was observed in all quadrants, where one of the signals kept differing, suggesting that signal overlap and interference occurred. Likely caused by obstacles within the room, it impacted the system's ability to detect quadrant boundaries accurately. Despite this, the experiment offered valuable insights into how signal interference can affect beacon-based localization systems. It highlighted the limitations of relying solely on BLE for precise position localization of assets within a room.

3.3.7. *UWB Localization Experiment - Multiliterate setup for Improved Localization Accuracy*

Experiment

Building on the findings from the quadrant experiment, this experiment aimed to improve inner room-level localization accuracy using two UWB modules. The experiment was conducted in Room 405-712, a square space with known corner coordinates. The two UWB modules were set at opposite diagonal corners, specifically at points (0, 0) and (9.45, 12.75) meters. The distance of these coordinates was measured by tape measure. The idea was to employ multilateration

techniques, leveraging the known distances between the UWB modules and the user's location to triangulate their precise position within the room. The UWB modules measured the distance to the user at specific reference points within the room. We used recorded distance values at key locations and compared these to theoretical values calculated using the Pythagorean theorem and simulated in Desmos.

Findings

The experiment demonstrated that placing two UWB modules at opposite corners of the room provided highly accurate localization. We first selected a position for comparison, with the actual user position measured at (5, 5.6). From this point, we recorded the distances: 7.3 meters from UWB module 1 at (0, 0) and 8.7 meters from UWB module 2 at (9.45, 12.75). Using the Pythagorean theorem in Desmos, we calculated the intersection point based on these distances, yielding coordinates of (5.11842, 5.20497), closely matching the measured position of (5, 5.6). This close alignment, with errors of less than 0.4 meters, indicates that the multilateration setup provided near-perfect accuracy for room-level localization. However, due to the use of only two UWB modules, there was a second intersection point at (3.49112, 6.41109). To address this, we considered averaging the two points to derive an approximate user location. While this method provided satisfactory accuracy for determining the vicinity of the asset's location within the room, additional UWB modules may be necessary in larger or more complex environments to ensure precise tracking and minimize potential blind spots.

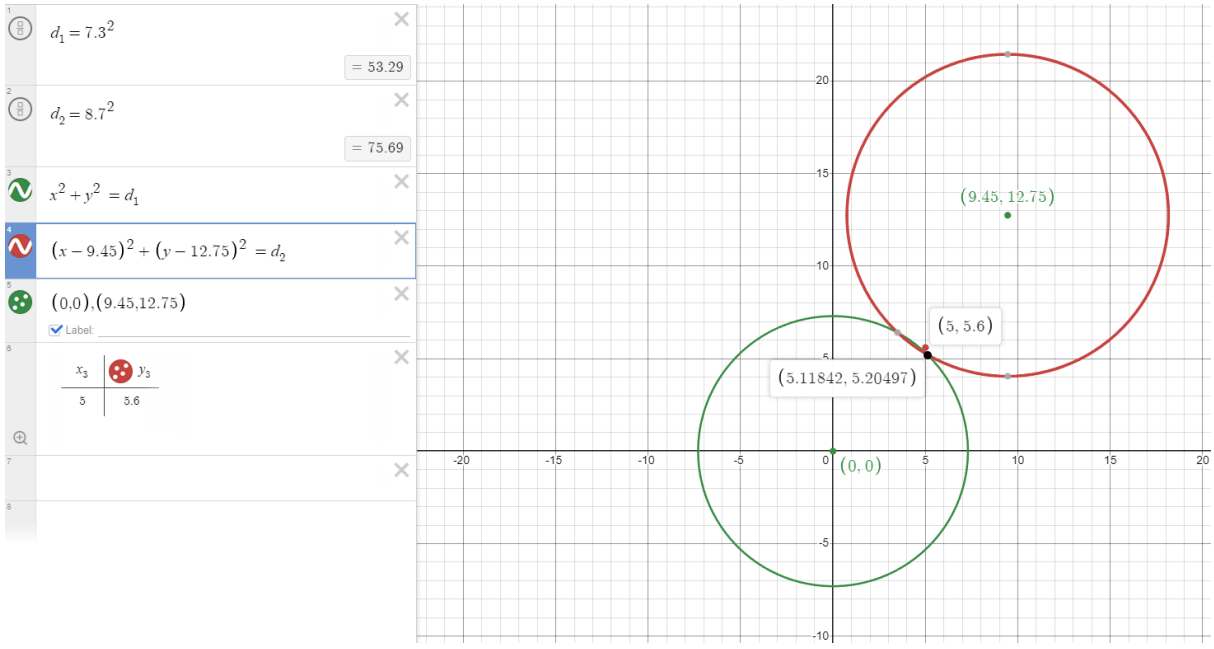


Figure 15: Inner-Room Localization Accuracy Test Using Multi-UWB Configuration.

3.4. BLE & UWB Hybrid Model

3.4.1. Theoretical Analysis of UWB and BLE hybrid model

In our earlier experiments with BLE, we demonstrated the feasibility of a smart indoor localization system, achieving a room-level detection accuracy of 91.7%. The main goal of our research, however, was to develop a smart asset-tracking system that goes beyond just identifying the room an asset is in but also tracks its precise location within that room. To address this, we propose a theoretical integration of Ultra-Wideband (UWB) technology, leveraging its precision, with the BLE-based system to refine tracking accuracy. Although, due to time limitations, we didn't have the opportunity to test this hybrid setup fully, our findings from previous experiments with BLE and UWB allow us to estimate the performance of such a system. The BLE beacons, configured as Four Wall-Mounted Beacons with an Auxiliary Beacon, would first detect room entry by measuring the strongest two signals to confirm the asset's location within the room. Once the asset enters the room and the UWB modules come within range, the system will use UWB's precise measurements (up to 0.1 meters) to determine the asset's exact location inside the room. A detailed visual representation of this configuration can be found in Appendix B, Figure B.10.

Based on our earlier data, we expect that this combined approach would significantly improve accuracy by compensating for the limitations of BLE's room-level detection and UWB's distance-based precision. For instance, UWB's multiliterate setup for high accuracy in detecting distance deviations (within 0.4 meters) with BLE's strong room-boundary identification maintains an accuracy of over 90%. By combining these technologies, we expect to mitigate room transition ambiguity and improve the effectiveness of the asset-tracking system for real-time inventory management. Though the hybrid setup has not been implemented across our system yet, the theoretical model suggests a promising future for further research and development. The dual-system approach, combining BLE's affordability and scalability with UWB's precision, offers a robust solution for smart asset management in university lab environments.

4. Results

This section presents the key findings from our indoor localization experiments and the performance of our prototype system. Two crucial elements were evaluated during the study: the optimal settings for BLE and UWB configuration setups and the overall effectiveness of room-level detection. Additionally, the prototype systems were also evaluated, including its ability to maintain real-time synchronization of asset locations and user data through our iOS app. The following sections provide a detailed analysis of the experiments and insights into the optimal settings for indoor localization using BLE and UWB beacons.

4.1. Overall Result

Our testing began by evaluating the range and accuracy of both BLE and UWB technologies. At a transmit power of +9 dBm, BLE demonstrated a longer range of 35-40 meters, while at a lower transmit power of -12 dBm, the range was reduced to 15-20 meters. UWB, in comparison, had a consistent range of 15-20 meters. In terms of accuracy, regression analysis showed that UWB maintained near-perfect accuracy across all environments, with $R^2 \approx 1$, indicating minimal deviation from theoretical values.

At a higher transmit power of +9 dBm, BLE's accuracy was weaker in open environments, with an R^2 of 0.385, but improved significantly in obstructed environments, reaching an R^2 of 0.855. It showed an R^2 of 0.83 in semi-obstructed environments, showing better stability in environments with more obstacles. This reflects BLE's ability to handle signal reflections and obstructions better at higher transmit power levels. In contrast, at a lower transmit power of -12 dBm, BLE performed more consistently in open spaces with an R^2 value of 0.71, but its range was reduced. In obstructed environments, the R^2 was slightly better at 0.713, while in semi-obstructed settings, it reached 0.773. Despite this, BLE's lower transmit power demonstrated better accuracy for short-range tracking, particularly in open spaces, where signal reflections are minimal. Overall, UWB proved to be a superior solution for high-accuracy indoor localization, and BLE with high transmission power is optimal for long-range coverage.

After the initial findings, we concentrated on improving room identification by analysing RSSI signal patterns and adjusting the number of beacons. Starting with two BLE beacons, we progressively increased the number to three, then five, to enhance room differentiation. The results indicated that increasing the density of beacons improved signal accuracy; however, relying solely on changes in RSSI patterns was insufficient for precise room-level detection. Therefore, our final experiment focused on RSSI measurement and optimizing beacon placement. Beacons positioned near walls rather than in the centre of the room were more effective in detecting room transitions by capturing stronger RSSI signals. Additionally, increasing beacon density in larger rooms further improved accuracy. This multi-beacon setup ultimately enhanced room transition detection and overall system performance.

The system achieved an overall room-level accuracy of 91.7%, with the app performing reliably at 100% availability across all time, ensuring dependable operation and system reliability. The optimal configuration identified was the *BLE Configuration Experiment – Four Beacons with Auxiliary Setup*, which surpassed the 77.1% accuracy obtained from the *Four Beacons for Enhanced Room Detection* setup.

4.2. System Architecture for Real-time Asset Tracking

Our system architecture integrates the BLE module (ESP32-H2) for indoor localization for asset

tracking in the University of Auckland labs. The BLE modules initially transmit RSSI signals, which then capture the two strongest RSSI signals, which are then processed by the Swift command line using conditional logic to select the two strongest signals, allowing accurate room-level positioning. Users log into the iOS app with their UPI and name and scan assets using QR codes, and the app locks the screen until the asset's location is updated in real time. The system uses Firebase Firestore to manage asset data to retrieve and update, ensuring real-time asset location and user information synchronization. Upon completing the asset movement, the user unscanned the asset across the same barcode and updated the location data displayed on the front-end UI. Full details of the system architecture, including diagrams and user flow interfaces, are provided in Appendix C.

5. Discussion

5.1. Evaluation of Research Objective

The primary goal of this research was to assess the feasibility and accuracy of room-level localization using BLE and UWB technologies. Although we mainly focused on BLE and were unable to fully implement a combined BLE and UWB solution, increasing BLE beacon density significantly improved RSSI accuracy and room boundary detection. The system's overall accuracy of 91.7% largely met our expectations, though it fell short of the near-perfect accuracy we initially aimed for.

5.2. Evaluation of Research Methodology

The methodology, adjusting beacon density and placement proved effective, with beacons placed at doorways yielding better room transition detection than central placements. While some challenges, such as signal degradation in complex layouts, were encountered, the study mostly achieved its objective of evaluating an optimal BLE configuration for accurate indoor localization.

5.3. Impact of Transmit power on Accuracy and Range of BLE

Our experiments demonstrated a clear relationship between transmission power and BLE's accuracy and range. When testing single BLE beacons, we observed that higher transmission power, such as 9 dBm, resulted in lower accuracy in open spaces but became more stable in environments with more obstacles. In contrast, lower transmission power (-12 dBm) showed more consistent accuracy in open spaces, but its accuracy decreased as the environment became more obstructed. Regarding range, high transmission power significantly extended BLE's range, reaching up to 35-40 meters, whereas lower transmission power limited the range to around 15-20 meters. This phenomenon aligns with findings from [31] and [32]. According to [31], lower transmission power reduces the issue of multipath propagation (where signals bounce off

surfaces like walls and floors), which can cause significant errors in location estimation, although the range is naturally limited. Lower signal strength makes it more susceptible to reflection effects, especially in obstructed environments. Meanwhile, [32] explains that higher transmission power reduces the effects of noise, maintaining more consistent signal strength over longer distances, which is beneficial in obstructed spaces, though at the cost of increased susceptibility to interference and instability in open environments.

5.4. Impact of BLE density on Accuracy

The density of BLE beacons plays a crucial role in enhancing localization accuracy. As observed in our experiments, increasing the number of beacons significantly improved the accuracy of RSSI measurements, which in turn improved the system's ability to detect room boundaries. This phenomenon aligns with the findings in [33], who noted that higher beacon density leads to improved estimation accuracy due to the increased availability of signal data points. Their study reported a notable reduction in estimation error as the number of beacons increased from four to eight, emphasizing the importance of multiple signal sources for reliable indoor positioning. Furthermore, when beacons are densely placed within the environment, the likelihood of capturing multiple RSSI signals from various angles increases, thereby reducing the impact of signal fluctuations caused by obstacles or interference. This redundancy allows the localization algorithm to better filter out noise and improve the reliability of position estimation. In our study, the transition from two to five beacons resulted in more accurate room-level detection, as well as the four to six beacons, reinforcing the importance of density in beacon configuration.

5.5. Impact of BLE configuration on Accuracy

The placement of BLE beacons significantly impacts localization performance. Our findings show that beacons positioned between walls provided better room transition detection than those placed centrally or at wall. As [33] emphasizes, the spatial arrangement of beacons greatly influences accuracy. They found that uneven or sparse placements led to increased localization errors, while a more systematic approach resulted in more precise estimations. In our experiments, placing beacons above doorways for clear line-of-sight resulted in stronger and more accurate signals, making it easier to detect room boundaries through stable RSSI pattern changes. When beacons are evenly distributed, coverage improves, reducing estimation errors in both position and direction. In contrast, central placements or those at one end of a wall often suffered from signal degradation, especially in larger or more complex layouts. The optimal configuration involved multiple wall-mounted beacons, minimizing interference and enhancing signal reliability.

6. Limitations and Future Work

6.1. Limitations

While this research demonstrated promising results in BLE-based indoor localization, several limitations were identified. First, although the system's accuracy improved with increased beacon density and configuration, accuracy was still away from achieving near-perfect accuracy we initially aimed for. Signal degradation, particularly in complex or obstacle-filled environments, remains a challenge. BLE alone may not provide the necessary precision for larger, more intricate spaces. Moreover, this study was limited to a single laboratory environment, restricting the system's evaluation to a specific layout and setup. The potential benefits of a hybrid BLE and UWB solution were not fully explored in this study. UWB's high precision, when combined with BLE's cost-effectiveness, could have addressed some of the accuracy challenges, especially in difficult environments. However, the incomplete implementation of the hybrid approach prevented a comprehensive assessment of how these technologies might complement each other to overcome signal degradation and improve accuracy across diverse environments.

6.2. Future Work

Future research should address these limitations by exploring the following areas:

- **Hybrid Approach**: Future work should focus on fully integrating BLE and UWB technologies into a hybrid system. BLE provides cost-effective general positioning, while UWB offers high precision, particularly in complex or obstructed environments. A hybrid system would enable BLE for broad coverage and UWB for high-accuracy scenarios, such as room transitions and dense environments. Studies like [34] suggest this dual-system approach effectively balances accuracy and cost-efficiency. By using BLE for general positioning of the room and UWB for its higher precision in multipath environments, performance could improve significantly over single-technology solutions.

- **Scalability Testing**: Another important area for future research is scalability. The current study was limited to a single laboratory environment on Floor 7 of the ECSE labs, restricting the understanding of how the system would perform in different or more complex settings. Future research should test the system across multiple laboratories or departments to evaluate its scalability and adaptability. This includes assessing how current beacon densities and placements affect accuracy in environments with varied layouts, materials, and interference sources. Testing in diverse spaces will help demonstrate the system's potential for broader deployment in real-world applications.

7. Conclusion

This research evaluated the feasibility of Bluetooth Low Energy (BLE) and Ultra-Wideband (UWB) technologies for indoor localization within a university lab environment. Our objective was to create a cost-effective and accurate system for real-time asset tracking. The system achieved an overall room-level detection accuracy of 91.7%, which was considered sufficient for this project's scope. However, there remains room for improvement to achieve near-perfect accuracy.

Our findings showed that increasing the density and changing the configuration of BLE beacons significantly improved accuracy, especially in detecting room transitions. Nevertheless, we identified challenges such as signal degradation in larger, more complex environments and the inherent limitations of BLE technology. The exploration of a hybrid approach combining BLE and UWB showed potential to address these issues, though we could not fully implement this within the project's timeframe.

Future research should explore the hybrid BLE-UWB system to realize its full potential for improving accuracy and efficiency in indoor localization. Expanding the system to other labs and environments could provide insights into its scalability and adaptability. In conclusion, this project demonstrated the feasibility of BLE for indoor localization, identified critical areas for improvement, and laid the groundwork for future work, particularly in combining BLE with UWB to achieve greater precision and scalability for asset tracking systems.

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Appendix A User Survey Design

In this appendix, the results of the User Survey responses are summarized. The survey gathered feedback on current asset management systems, focusing on user experience, system efficiency, and expectations for improvements in the new system.

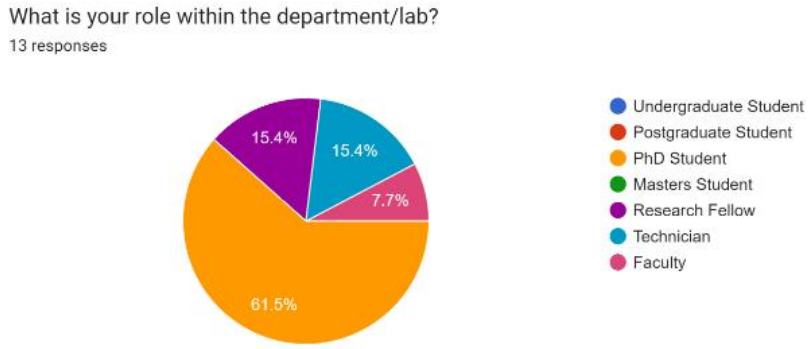


Figure A1: Demographic Distribution of Survey Participants in the Department/Lab

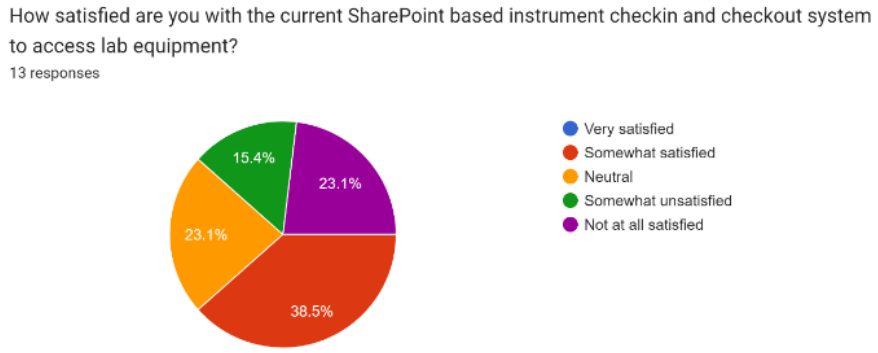


Figure A2: User Satisfaction with Current Check-In/Check-Out System

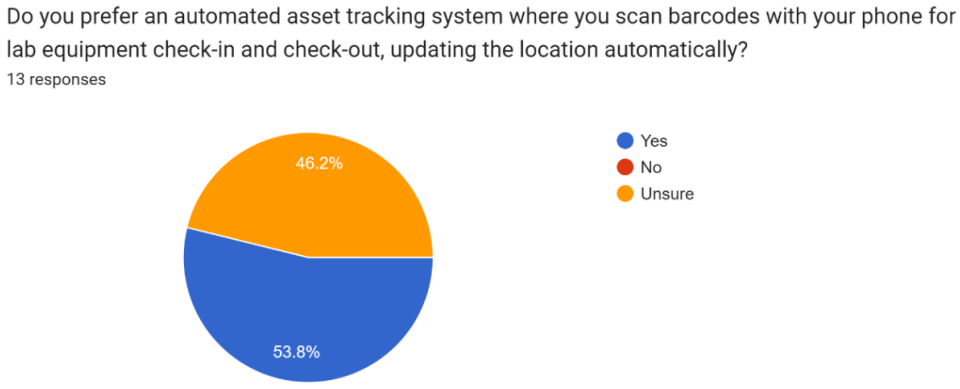


Figure A3: User Preference for Mobile Barcode Asset Tracking System

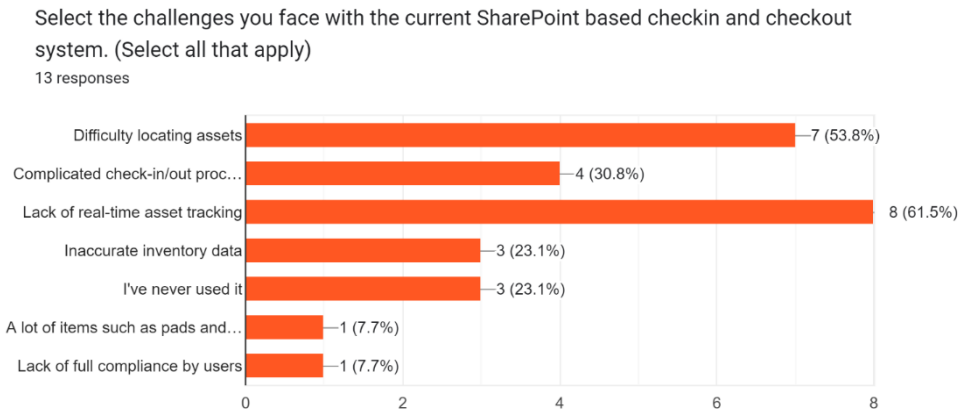


Figure A4: User Survey on Challenges with the Current Check-In/Check-Out System

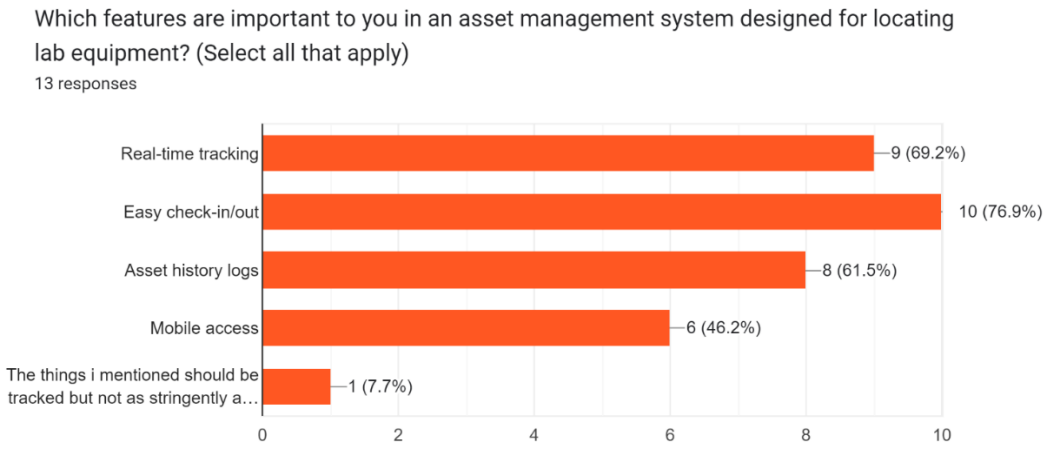


Figure A5: User Survey on Desired Features for an Asset Management System

Appendix B Experimental Tests and Configurations

This appendix provides the experimental setup, procedures, and accompanying photographs for all tests conducted throughout the study. While images from the closed environment in lab 405-636 were not captured due to restricted access, detailed descriptions of the test configurations, procedures, and results for each experiment are included below.

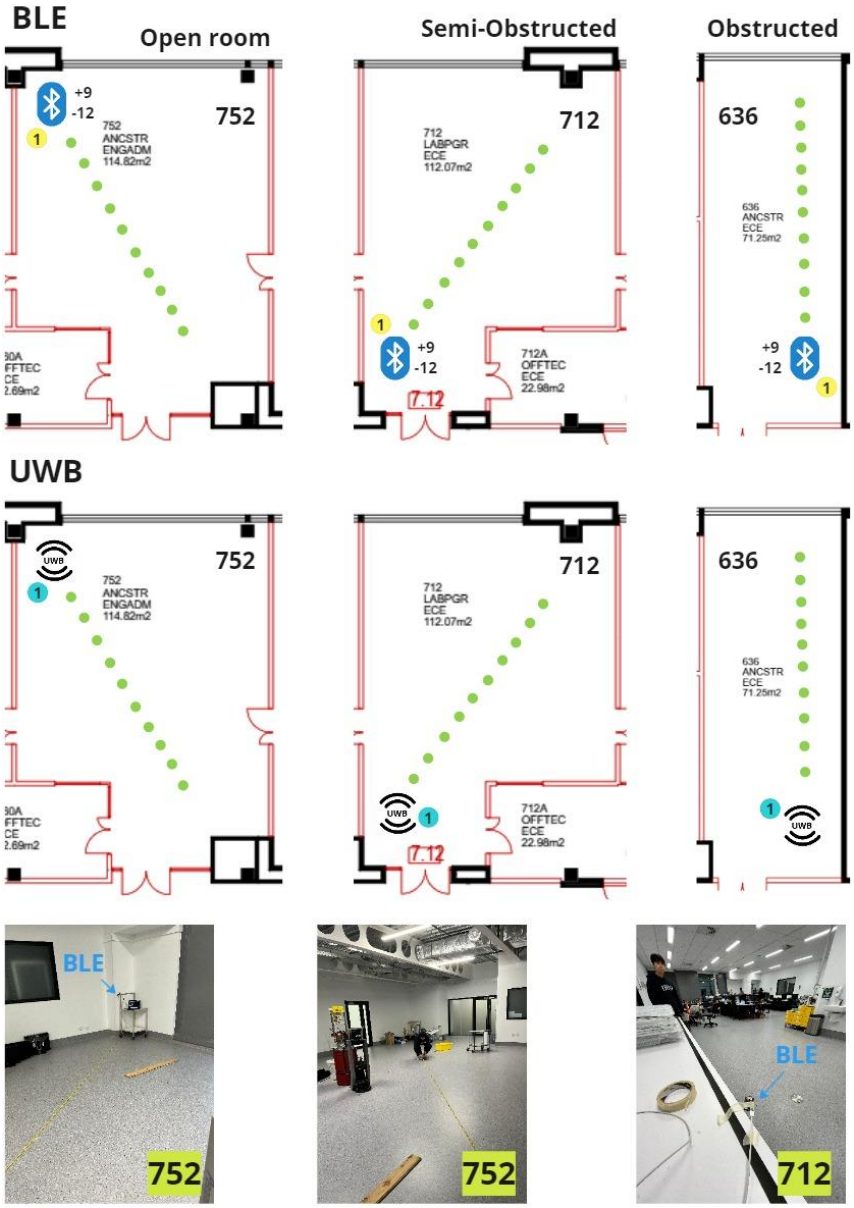


Figure B1: Photographs of the Experimental Setup for BLE and UWB Accuracy Testing in Lab 405-752 (Open), 405-712 (Semi-Obstructed), and 405-636 (Obstructed)

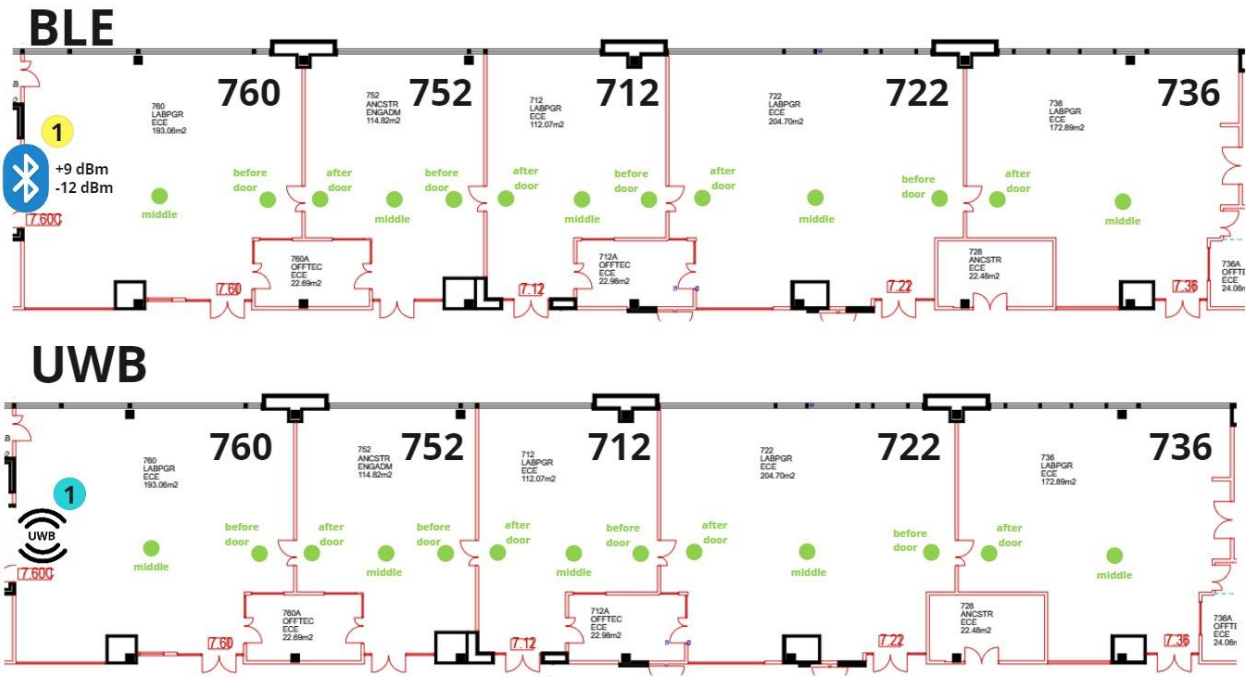


Figure B2: Photographs of the Experimental Setup for BLE and UWB Range Testing across Floor 7

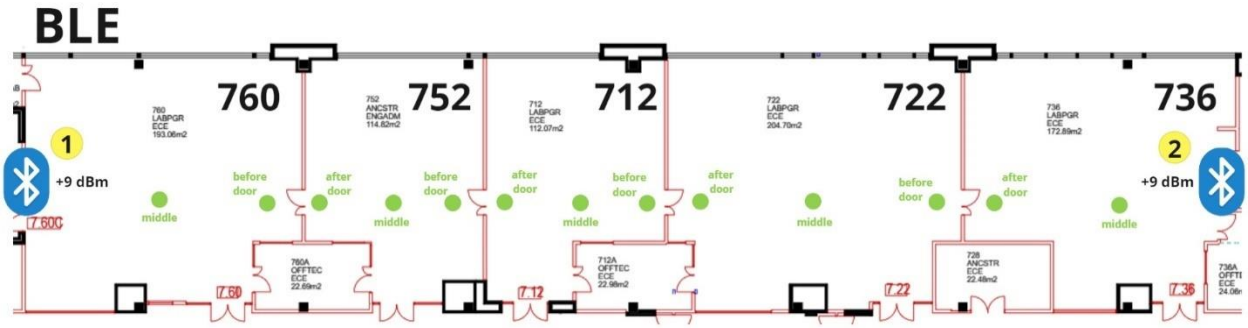


Figure B3: Visuals of the RSSI Pattern Experiment Using Two BLE Beacons

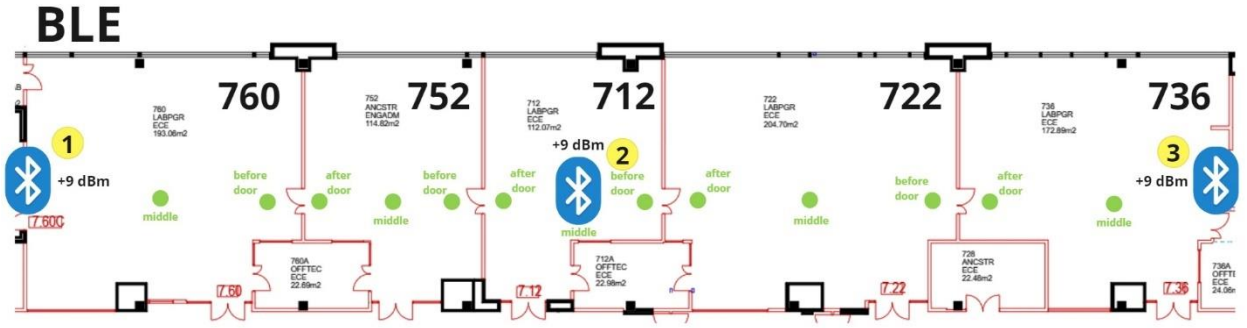


Figure B4: Visuals of the RSSI Pattern Experiment Using Three BLE Beacons

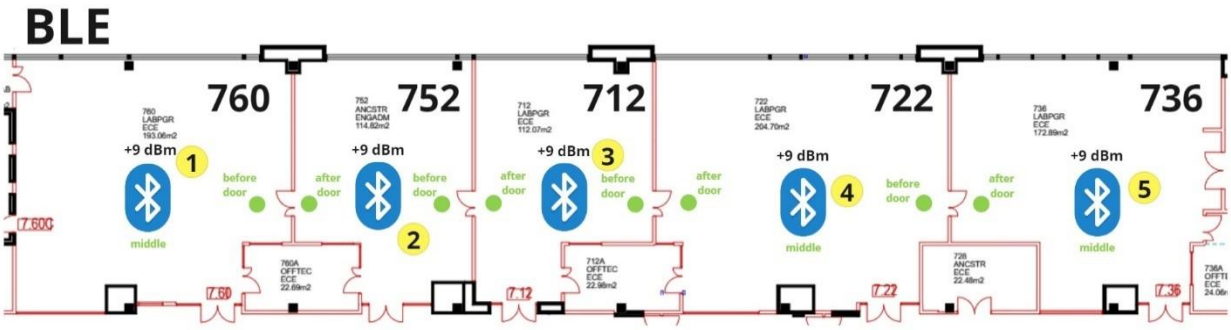


Figure B5: Visuals of the RSSI Pattern Experiment Using Five BLE Beacons

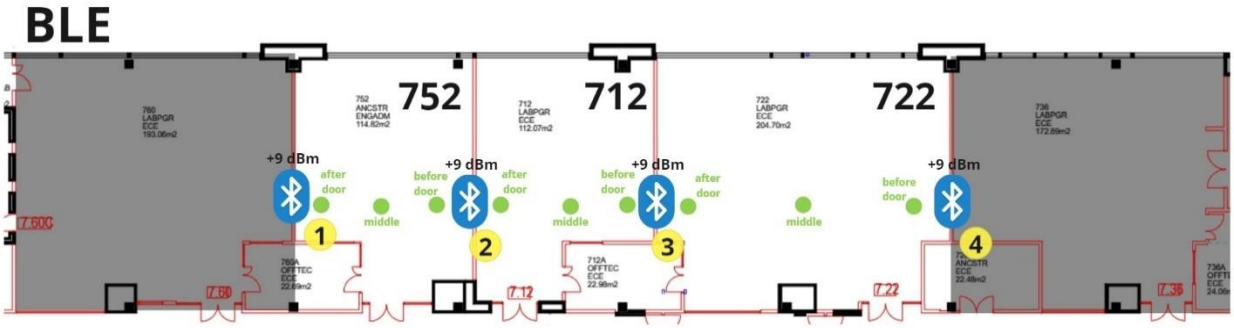


Figure B6: Layout of Four BLE Beacon Configuration Across Multiple Rooms for Improved Localization Accuracy

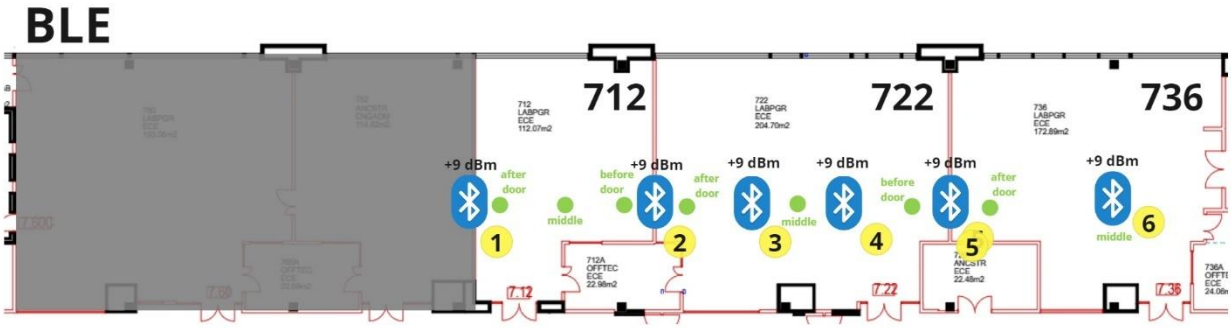


Figure B7: Layout of Six BLE Beacon Configuration Across Multiple Rooms for Improved Localization Accuracy

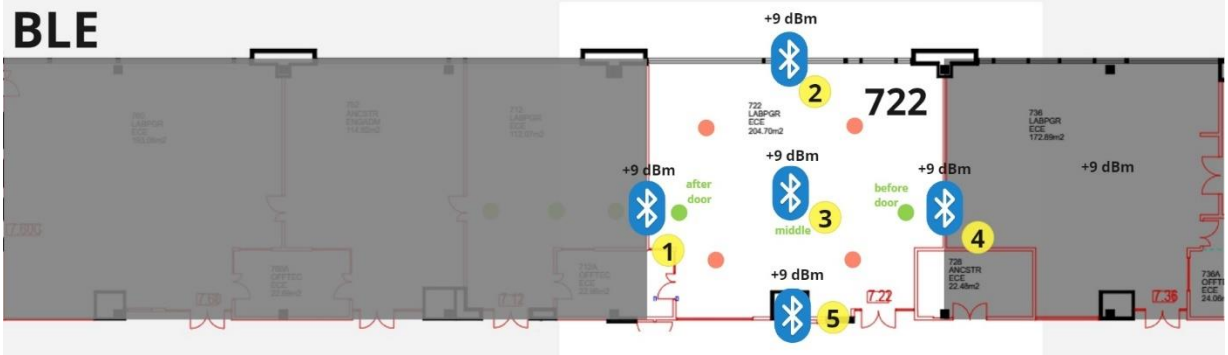


Figure B8: Layout of the BLE localization Experiment for Five BLE

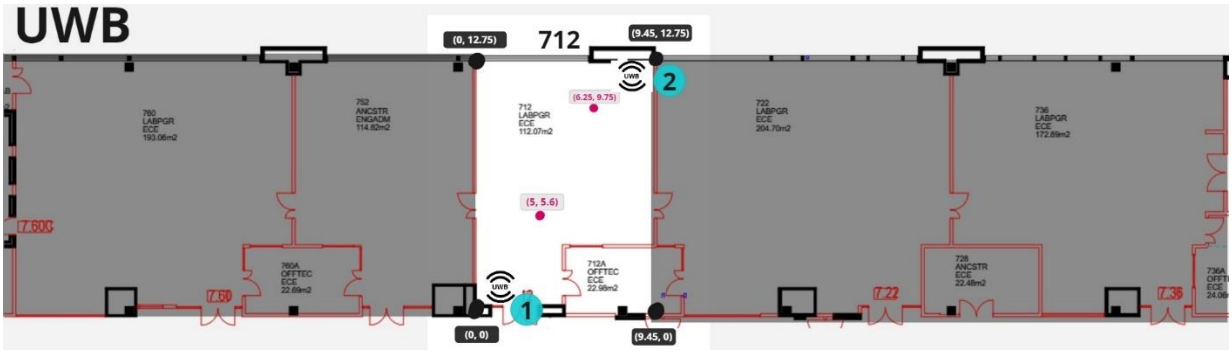


Figure B9: Layout of the UWB localization Experiment for two UWB

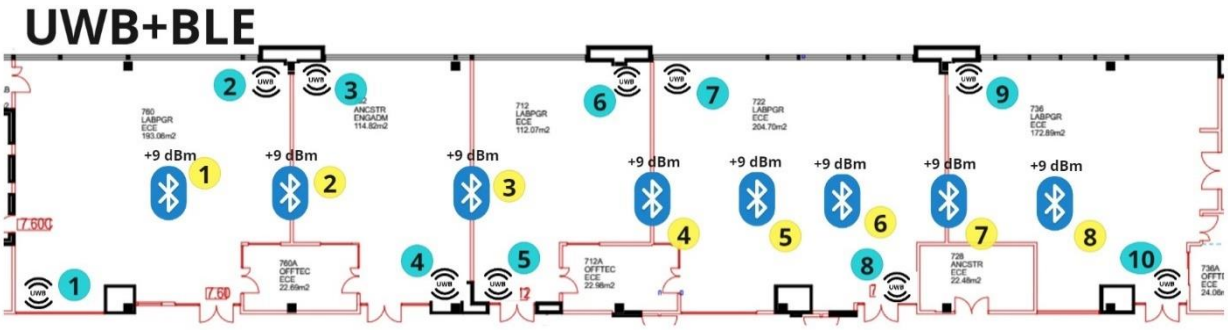


Figure B10: Visual Representation of Theoretical Hybrid Model for UWB and BLE Systems

Appendix C System Architecture

In this appendix, it provides the system architecture, user flow diagrams, and application UI designs. These visuals illustrate the structure and functionality of the asset management system, demonstrating how the components of the system interact.

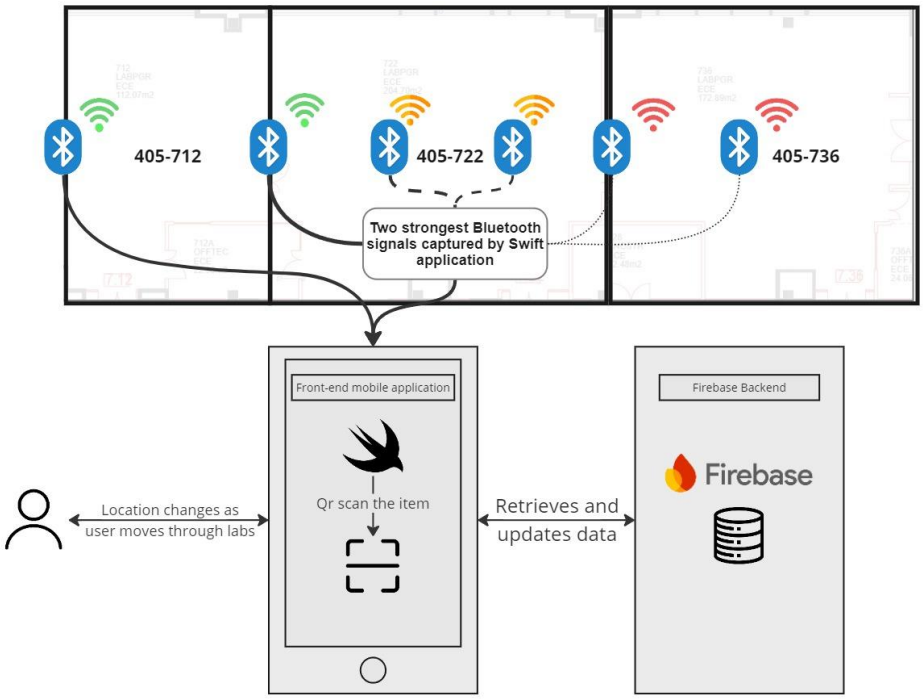


Figure C1: Overall System Architecture

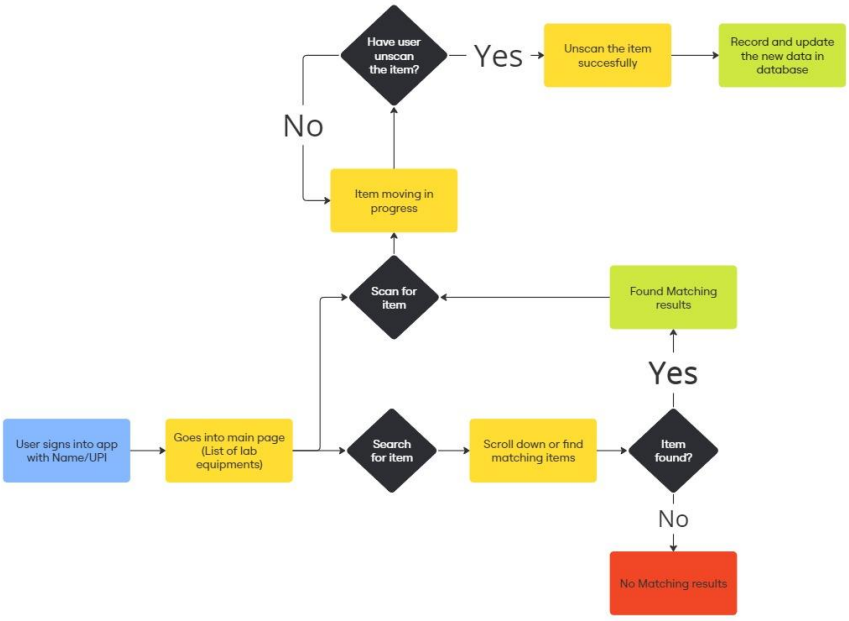


Figure C2: User Flow Diagram

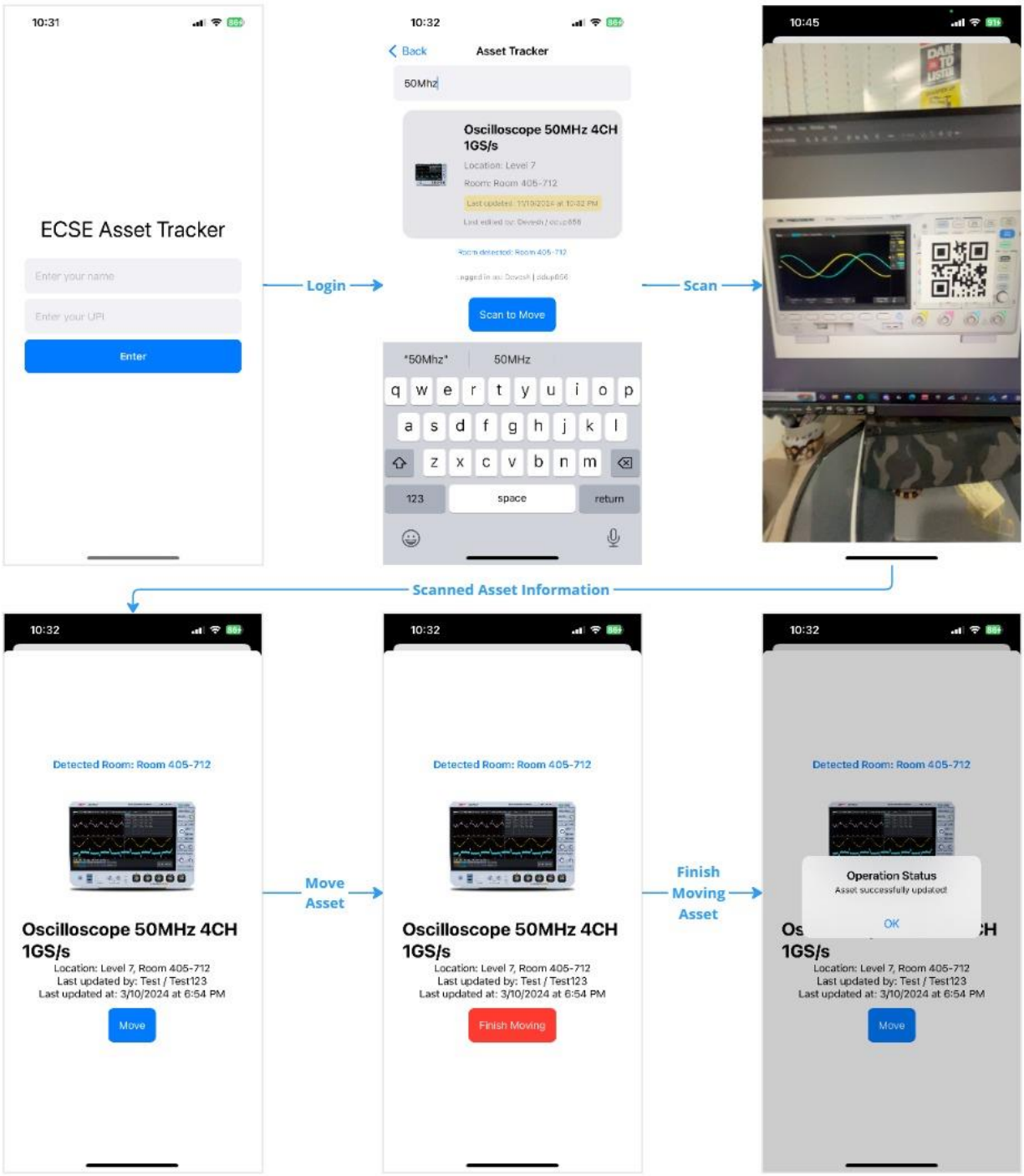


Figure C3: User Interface for Asset Tracker Application

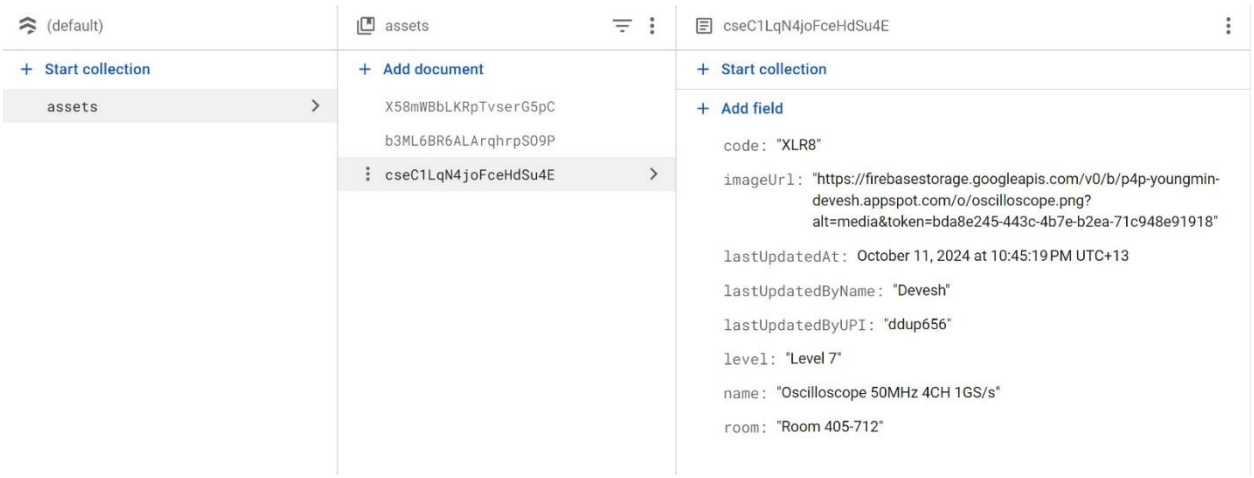


Figure C4: Database Schema